

$$\tilde{K}_v(u,0) = 4 (L\tilde{N})_v(u,0)$$

$$\begin{cases} L\tilde{N}(u,0) = 0 \dots ① \\ L\tilde{N}_v(u,0) = \frac{1}{12} \Pi_{3,5}(u) \dots ② \end{cases}$$

① を因子の補題で書き直すと $\exists n_1$ s.t. $\underline{L\tilde{N}(u,v) = V n_1(u,v)} \quad ③$

$$f, \bar{f} : ds_f^2 = \varphi^* ds_{\bar{f}}^2$$

$$\boxed{\begin{aligned} f \text{ が } \tilde{K}_v(u) \equiv 0 \text{ を } \partial T = \partial \cdot \\ \Rightarrow \bar{f} \text{ も } \tilde{K}_{\bar{v}}(u) = 0 \end{aligned}}$$

← "chp" 成り立ち

$\Pi_{3,5}$ is intrinsic

$$\Pi_{3,4} = L(u,0) \tilde{N}(u,0) \quad L(u,0) = \tilde{K}_v(u)$$

$$= \tilde{K}_v(u) \omega_{3,4}(u)$$

$$= \tilde{K}_{\bar{v}}(u) \bar{\omega}_{3,4}(u) = \bar{\Pi}_{3,4}(u)$$

$$\text{t.l. } \tilde{K}_v(u) = 0 \Rightarrow \tilde{K}_{\bar{v}}(u) \bar{\omega}_{3,4}(u) \equiv 0 \Rightarrow \tilde{K}_{\bar{v}}(u) = 0$$

$$\begin{aligned} \tilde{K}_{(u,v)} &= \frac{L\tilde{N} - V^2 \tilde{M}^2}{E\tilde{G} - \tilde{F}^2} \quad \text{③ 代入} \\ &= V \frac{n_1 - V \tilde{M}^2}{E\tilde{G} - \tilde{F}^2} (u,v) \end{aligned}$$

$$\tilde{K}_v(u,v) = \frac{n_1 - V \tilde{M}^2}{E\tilde{G} - \tilde{F}^2} + V \left(\quad \right)_v$$

$$\tilde{K}_v(u,0) = \frac{n_1}{(E\tilde{G} - \tilde{F}^2)(u,0)}$$

$$= 4 n_1(u,0) \quad \text{③ より}$$

$$= 4 (L\tilde{N})_v(u,0)$$

$$= 4 \left(\underline{L_v \tilde{N}(u,0)} + L \tilde{N}_v(u,0) \right)$$

$$= 4 \left(\underline{L_v(u,0)} \frac{1}{6} \omega_{3,4}(u) + \frac{1}{12} \Pi_{3,5}(u) \right)$$

(3.4) - cuspidal edge

= 3-type edge + wave front



degenerate order=2

1st Kind

∴ 3-type edge は 内在的に判定可

だが wave front は判定不可

Ex (3.4)-CE, (3.5)-CE : 3-type edge

内在的な不変量 $Th.4, Th.5$ をみつけたい

応用 → (3.4)-CE, (3.5)-CE の 内在的な判定条件
を与えたい

判定条件 $f: U \rightarrow \mathbb{R}^3$: 7D-VAL

$p \in U$: 特異点 とする

(i) $f: (U, p) \rightarrow \mathbb{R}^3$: (3.4)-CE と A-同値

$\iff$ (i) f は 3-type edge $\iff f$ は 3-type edge
(ii) f は 7D-VAL $Wh.4(p) \neq 0$

内在的な不変量 $\pi_{3,4}$, $\pi_{3,5}$ を与えたい.

応用 $\rightarrow$ (3,4)-CE, (3,5)-CE の内在的な判定条件
を与えたい.

判定条件 $f: U \rightarrow \mathbb{R}^3$: generic 7点

$p \in U$: 特異点 とする.

(i) $f: (U, p) \rightarrow \mathbb{R}^3$: (3,4)-CE と A -B 通過

$\iff$ (i) f は 3-type edge $\iff f$ は 3-type edge
(ii) f は 7点 $\iff \omega_{3,4}(p) \neq 0$

(i) のとき

(ii) $\iff \omega_{3,4}(p) \neq 0$

内在的な判定条件 $f: U \rightarrow \mathbb{R}^3$: generic 7点

(i)' $f: (U, p) \rightarrow \mathbb{R}^3$ は (3,4)-CE と

$\iff f$ は 3-type edge

$\pi_{3,4}(p) \neq 0$

or $f, \bar{f}$: generic 7点

$f, \bar{f}$ は 等長 である.

条件

(i) のとき

$$(ii) \iff W_{3,4}(p) \neq 0$$

内在的判定条件 $f: U \rightarrow \mathbb{R}^3$: generic 78
702711.

$$(1)' f: (U, p) \rightarrow \mathbb{R}^3 \text{ は } (3,4)\text{-LE} \text{ と } A \text{ 同値}$$

$$\iff f \text{ は } 3\text{-type edge}$$

$$\Pi_{3,4}(p) \neq 0$$

or $f, \bar{f}$: generic 78
 $f, \bar{f}$ は等長的と証明

$$f: (U, p) \rightarrow \mathbb{R}^3 \sim_{\lambda} (3,4)\text{-LE}$$
$$\Downarrow$$
$$\bar{f}: (U, p) \rightarrow \mathbb{R}^3 \sim_{\lambda} (3,4)\text{-LE}$$

$$(2) f \text{ at } p \sim_{\lambda} (3,5)\text{-LE}$$

$$\iff (i) f \text{ は } 3\text{-type edge}$$

$$(ii) f \text{ は } \gamma(H) \text{ 上で } (3,5)\text{-LE}$$

$(\iff W_{3,4}(u) \equiv 0)$ ではない

$$(i') W_{3,5}(p) \neq 0$$

$$(2)' f: \text{generic のとき}$$

$$f \text{ at } p \sim_{\lambda} (3,5)\text{-LE} \iff$$

$$(i) 3\text{-type edge}$$

$$(ii) \Pi_{3,4}(p) \equiv 0$$

$$(iii) \Pi_{3,5}(p) \neq 0$$

おけたい

判定条件

同値

と等しい

-(EとA)同値

$\Leftrightarrow f$ is 3-type edge
 $w_{3,4}(p) \neq 0$

(i) のとき

$$(ii) \Leftrightarrow w_{3,4}(p) \neq 0$$

判定条件 $f: U \rightarrow \mathbb{R}^3$: generic 78
70777

(1)' $f: (U, p) \rightarrow \mathbb{R}^3$ が (3.4)-LE と A 同値

$\Leftrightarrow f$ is 3-type edge

$$\pi_{3,4}(p) \neq 0$$

or $f, \bar{f}$: generic 70777
 $f, \bar{f}$ は等長のと等しい

$$(2) f \text{ at } p \sim_{\lambda} (3,5)\text{-LE}$$

$$\Leftrightarrow (i) f \text{ is 3-type edge}$$

$$(ii) \underbrace{f \text{ is } \gamma(H) \text{ 上 } \gamma(H)}_{(\Leftrightarrow w_{3,4}(u) \equiv 0) \text{ 27777}}$$

$$(iii) w_{3,5}(p) \neq 0$$

$$(2)' f: \text{generic のとき}$$

$$f \text{ at } p \sim_{\lambda} (3,5)\text{-LE} \Leftrightarrow$$

- (i) 3-type edge
- (ii) $\pi_{3,4}(u) \equiv 0$
- (iii) $\pi_{3,5}(p) \neq 0$

$$f: (U, p) \rightarrow \mathbb{R}^3 \sim_{\lambda} (3,4)\text{-LE}$$

$$\Downarrow$$
$$\bar{f}: (U, p) \rightarrow \mathbb{R}^3 \sim_{\lambda} (3,4)\text{-LE}$$

内在的判定条件

$f: U \rightarrow \mathbb{R}^3$: generic Toric.
 $p \in U$ is a pt.

(2)' f at $p \sim_{\mathcal{A}} (3.5)$ -CE

$\Leftrightarrow \begin{cases} (i) f \text{ is 3-type edge at } p. \\ (ii)' \pi_{34}(q(p)) = 0. \\ (iii)' \pi_{35}(p) \neq 0. \end{cases}$

問1 $f, \bar{f}$: generic Toric.
 $f, \bar{f}$: 等長的 arc.

f at $p \sim (3.5)$ -CE $\Leftrightarrow \bar{f}$ at $p \sim (3.5)$ -CE

$\pi_{34}(q(p)) = 0 \Rightarrow \pi_{35}(p)$: intrinsic test. OK

is a pt. $\uparrow$ is the point of the arc.

f at p (i), (ii)', (iii)' is a pt.

$\Rightarrow \bar{f}$ is (i), (ii)' is a pt. (π_{34} of the arc is 0)

is a pt. (iii)' is a pt. is not clear.

(3.4)-CE? $K_V = 0$ is $L_V(u,0) \neq 0$ of the arc

問2 問1の反例を作れるか?

is a pt. $f, \bar{f}$: generic Toric. 等長的 arc

f is (3.5)-CE is a pt.

$\bar{f}$ is (3.5)-CE is not clear.

問3 $\pi_{34} \equiv \pi_{35} = 0$ is a pt. K : the arc is a pt.

$\tilde{K}_V(u,0) = L_V(u,0)N(u,0) + \pi_{35}(u)$
 $\uparrow$
 $\pi_{34}(u) = 0$ at a pt.
 $\tilde{K}_V(u) = 0$ is a pt. $\pi_{34}(u) = \pi_{35}(u) = 0$.

内在的判定条件

$f: U \rightarrow \mathbb{R}^3$: generic Toric.

$p \in U$ is a pt.

(2)' f at $p \sim (3.5)$ -CE

$\Leftrightarrow \begin{cases} (i) f \text{ is 3-type edge at } p. \\ (ii)' \Pi_{2,4}(q) = 0. \\ (iii)' \Pi_{2,5}(p) \neq 0. \end{cases}$

問1 $f, \bar{f}$: generic Toric.

$f, \bar{f}$: 等長的 at p .

f at $p \sim (3.5)$ -CE $\Leftrightarrow \bar{f}$ at $p \sim (3.5)$ -CE

$\Pi_{2,4}(q) = 0 \Rightarrow \Pi_{2,5}(p)$: intrinsic is OK.

is pt. $\uparrow$ is the point to be shown.

f at p : (i), (ii)', (iii)' is OK.

$\Rightarrow \bar{f}$ is (i), (ii)' is OK. ($\Pi_{2,4}$ of intrinsic is OK)

but (iii)' is OK is not clear.

(3.4)-CE is $K_v = 0$ or $L_v(u,0) \neq 0$ of (3.1)

問2 問1の反例を作れるか?

例. $f, \bar{f}$: generic Toric. 等長的 at p .

f is (3.5)-CE is OK.

$\bar{f}$ is (3.5)-CE is not clear.

問3 $\Pi_{2,4} = \Pi_{2,5} = 0$ is OK? K_v is not zero is OK?

$\tilde{K}_v(u,0) = L_v(u,0) \tilde{N}_v(u,0) + \Pi_{2,5}(u)$
 $\uparrow$
 $\Pi_{2,4}(u) = 0$ at p .
 $\tilde{K}_v(u) = 0$ for S . $\Pi_{2,4}(u) = \Pi_{2,5}(u) = 0$.

内在的判断条件

$$f: U \rightarrow \mathbb{R}^3: \text{generic Toric.}$$
$$p \in U \quad 1 = 77L.$$

(2)' f at p  (3.5) - CE

$$\Leftrightarrow \text{(ii) } f \text{ is 3-type edge at } p.$$
$$(ii)' \quad \Pi_{3,4}(\gamma(t)) \equiv 0.$$
$$(iii)' \quad \prod_{25} (p) \neq 0.$$

Def 1. $f, \bar{f}$: generic 70%
in E of art

$f, \bar{f}$: 等長の文字.

$$f \text{ at } p \sim (3.5) - \text{CE} \iff \bar{f} \text{ at } p \sim (3.5) - \text{CE}$$

HW

作るべき例

$f, \bar{f} : K \neq 0$
70% 72.

f, f: 等長的.

$$\pi_{3,4}^f(\gamma_H) \equiv 0$$
$$\prod_{3 \leq j} f_j(p) \neq 0.$$
$$\pi_{2,4}^F(x+1) \equiv 0$$
$$\pi_{2.5}^F(p) = 0$$
$$\Pi_{3,4}(\eta) \equiv 0 \Rightarrow \Pi_{2,5}(p) : \text{intrinsic IFS. OK}$$

たが、 $\uparrow$ は現時点で示せぬ

f. pl. (i), (ii), (iii) $\Sigma \alpha_T = 9$

$\Rightarrow \bar{f}$ は (i), (ii)' を満たす. ($\Gamma_{3,4}$ の内点列)

$L \vdash L$. (iii)' $\Sigma \vdash \neg p$ 不明.

HW

(2.47) - CEZ.
 $K_V \equiv 0$ 2' $L_V(u,0) \neq 0$
 | の 1311

問2 問1の反例を作れか？

781. f.f : generic 70-71. 第 7 卷的 21

$f_{10}(3.5) = 0.75$

$$\bar{f}(18, 3.5) - \frac{(\bar{f}'_{x_1})_{(18, 3.5)}}{(\tilde{K}_{x_1})_{(18, 3.5)}}$$

明 3 Th. 4 = 1111
有果吃... 1111?

K: 有界闭集

$E = \langle \mathbf{p} |$
 $(E - \tilde{H}_0)$
 $\tilde{K}_V(u) = L_V(u) \tilde{N}(u) + T_{V,1}(u)$
 $\tilde{K}_V(u) = 0$
 $T_{V,1}(u) = T_{V,1}(u) = 0$

$\uparrow$
 $\pi_{34}(u) \equiv 0 \text{ a.e.}^\#$
 $\pi_{44}(u) \equiv 0 \text{ a.e.}^\#$

(iv) adapted coord g_i ($\partial/\partial x_i$)

$$\iff (1) \nabla g_i = (dx_i)$$

$$(2) f_p(u, s) = f_u(u, s) = 0$$

$$(3) \{ f_p(u, s), f_{u_i}(u, s), \gamma(u, s) \}$$

orthonormal frame

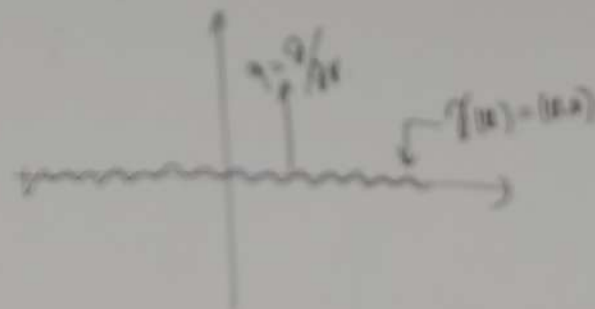
$$x \text{ long } f(x, s) = \langle x, s \rangle$$



$$\iff \begin{cases} E(u, s) = 1 \\ F(u, s) = F_x(u, s) = F_{u_i}(u, s) = 0 \\ G(u, s) = G_u(u, s) = G_{u_i}(u, s) = G_{u_i u_j}(u, s) = 0 \\ G_{u_i u_j}(u, s) = 0 \end{cases}$$

(iv) orthogonal coord.

$$\iff \begin{cases} E > 0 \\ F(u, s) \equiv 0 \\ G(u, s) = 0 \end{cases}$$



(2.0) adapted coord γ_s (AC)

$\Leftrightarrow$ (1) $\gamma(u) = (u, 0)$

(2) $\int_{\gamma} \langle u, v \rangle = \int_{\gamma} \langle u, v \rangle = 0$

(3) $\left\{ \int_{\gamma} \langle u, v \rangle, \int_{\gamma} \langle u, w \rangle, \gamma(u, v) \right\}$

orthonormal frame

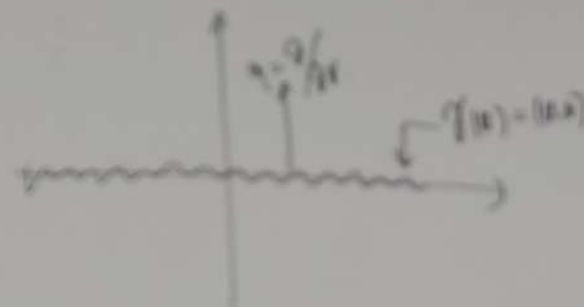
along $\gamma(u) = (u, 0)$



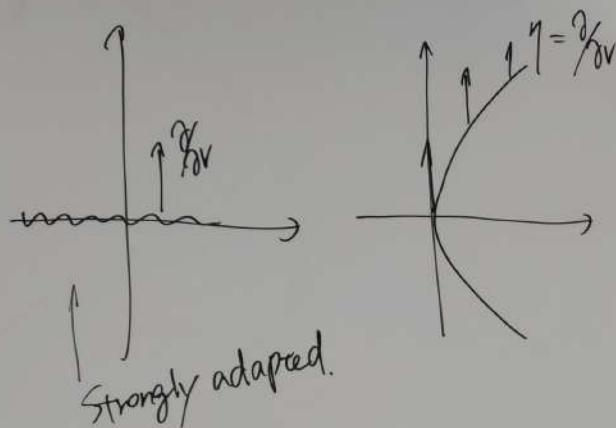
$\Leftrightarrow \begin{cases} E(u, v) = 1 \\ F(u, v) = F_x(u, v) = F_{xy}(u, v) = 0 \\ G(u, v) = G_x(u, v) = G_{xy}(u, v) = G_{yy}(u, v) = 0 \\ G_{yy}(u, v) = 0 \end{cases}$

(4.0) orthogonal coord.

$\Leftrightarrow \begin{cases} E > 0 \\ F(u, v) \equiv 0 \\ G(u, v) = 0 \end{cases}$



$$\text{adapted} \iff \eta = \frac{\partial}{\partial v} : \text{null}$$



$$\tilde{K} = v^2 K.$$

$$\Pi_{3,4}(u) := \tilde{K}(u, 0)$$

$$\Pi_{3,5}(u) := \tilde{K}_v(u, 0)$$

$$(\xi, \eta) \text{ is } g = \tilde{E} d\xi^2 + 2\tilde{F} d\xi d\eta + \tilde{G} d\eta^2$$

$$(1) \tilde{E}(\xi, 0) = 1$$

$$(2) \tilde{F}(\xi, \eta) \equiv 0.$$

$$(3) \tilde{G}(\xi, 0) = \tilde{G}_\eta(\xi, 0) = 0 \\ \tilde{G}_{\eta\eta}(\xi, 0) \neq 0.$$

Thm A $\Pi_{3,4}$: intrinsic inv.

$$\Pi_{3,4} > 0 \Rightarrow K \rightarrow +\infty$$

$$\Pi_{3,4} < 0 \Rightarrow K \rightarrow -\infty$$

$$\text{Thm B } \Pi_{3,4} > 0 \Rightarrow f \text{ is convex}$$

$$\Pi_{3,4} < 0 \Rightarrow f \text{ is concave}$$

Thm C (3,4)-CE of判定条件.

Con D $f, \bar{f}$: generic, homeric
or not $f : (3,4)\text{-CE} \iff \bar{f} : (3,4)\text{-CE}$

K : 有界
の条件不明

700ページ 2 n -R Kossowski 計量 と m -type edge
($m = n+1$)

- f : 退化双数 n 特異点
- f : 退化双数 n , 第一種
 $\iff f$: $(n+1)$ -type edge
- 特異曲率 K_s (偶奇で定義が違う)

Gauss-Bonnet 型 公式 (偶奇で)

$$n: \text{奇} \Rightarrow \begin{cases} \frac{1}{2\pi} \int K d\hat{A} = \chi(M_+) - \chi(M_-) + \sum_{p: \text{第2種}} \text{Ind}(p) \\ \frac{1}{2\pi} \int K dA + \frac{1}{\pi} \int K_s ds = \chi(M) \end{cases}$$

$$n: \text{偶} \Rightarrow \frac{1}{2\pi} \int K dA = \chi(M)$$

$$K_{\text{sc}} = \begin{cases} -\infty & (n=1) \\ \text{有界} & (n \geq 2) \end{cases}$$

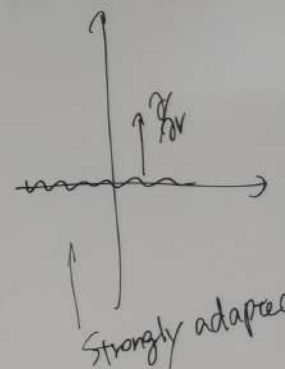
n -R Kossowski 計量 の定義

(Thm 連接境界の存在証明)

Thm 計量の等価定理

問. g : n -R Koss. 計量
 $p \in M^2$: 第一種特異点
 $\Rightarrow$ $\exists$ normalized s.a.c.s.

adapted $\iff$



$$\tilde{K} = v$$

Th. 4 (1)

Th. 5

(n=1)

(n ≥ 2)

量の定義

存在証明)

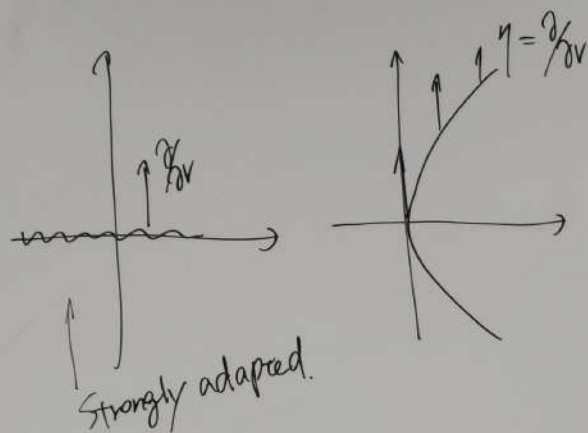
等価表現

g : n-次 Koss. 計量

$p \in M^2$: 第一種特異点

$\exists$? normalized s.a.c.s.

adapted $\Leftrightarrow \eta = \frac{\partial}{\partial v}$: null



$$\tilde{K} = v^2 K.$$

$$\pi_{2,4}(u) := \tilde{K}(u, 0)$$

$$\pi_{1,5}(u) := \tilde{K}_v(u, 0)$$

(ξ, η) は $g = \tilde{E} d\xi^2 + 2\tilde{F} d\xi d\eta + \tilde{G} d\eta^2$

(1) $\tilde{E}(\xi, 0) = 1$

(2) $\tilde{F}(\xi, \eta) \equiv 0.$

(3) $\tilde{G}(\xi, 0) = \tilde{G}_\eta(\xi, 0) = 0$
 $\tilde{G}_{\eta\eta}(\xi, 0) \neq 0.$

Thm A $\pi_{3,4}$: intrinsic inv.

$$\pi_{3,4} > 0 \Rightarrow K \rightarrow +\infty$$

$$\pi_{3,4} < 0 \Rightarrow K \rightarrow -\infty$$

Thm B $\pi_{3,4} > 0 \Rightarrow f$ は凸

$$\pi_{3,4} < 0 \Rightarrow f$$
 は凹状

Thm C (3,4)-CE の判定条件.
 $f, \bar{f}$: generic, isometric

Con D

$$f: (3,4)\text{-CE} \Leftrightarrow \bar{f}: (3,4)\text{-CE}$$

K : 有界
の条件は不明